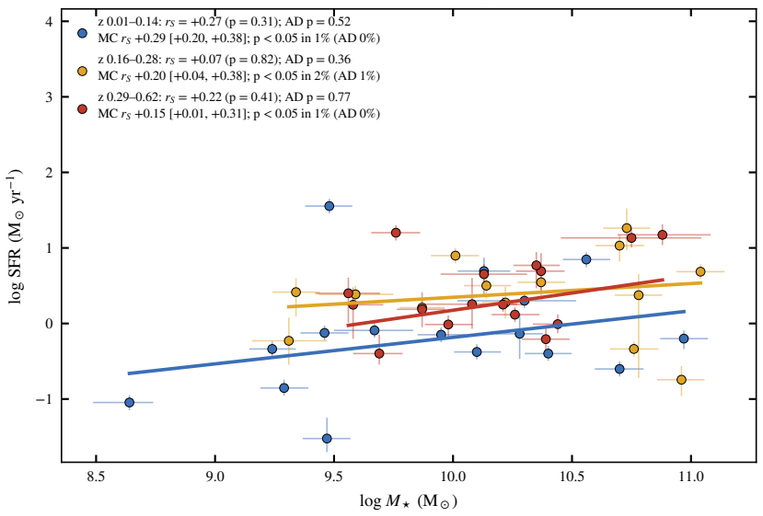
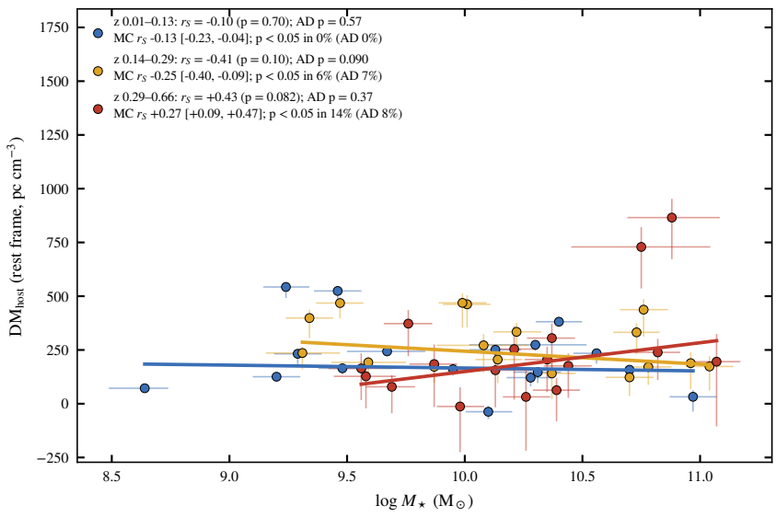
# Pairs of z, M*, SFR and DM_host within tertiles of the fourth quantity

## SFR vs M*, binned by z



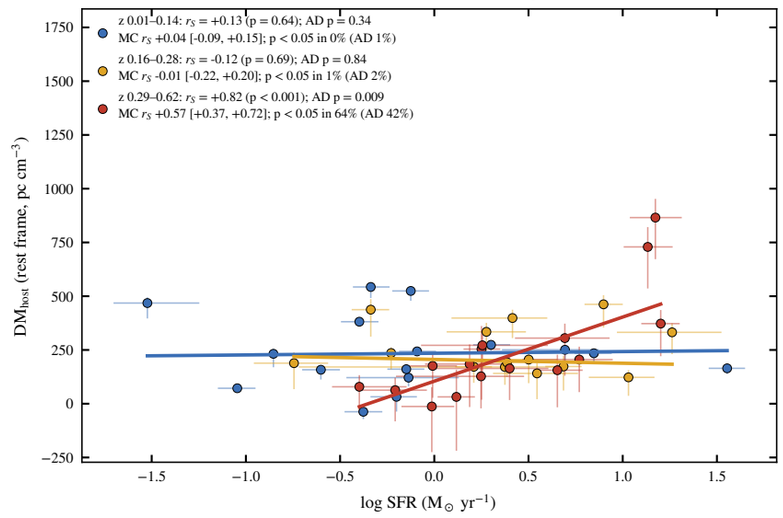

all 46: $r_S$ = +0.25 (p = 0.097); AD p = 0.43; partial $\tau$ | z = +0.15 (p = 0.11)
pooled within tertiles: $r$ = +0.20 (p = 0.24); tertiles agree: p = 0.88
through the errors (MC median [16, 84]%): all $r_S$ +0.23 [+0.17, +0.28], p < 0.05 in 13% (AD 1%)
MC partial $\tau$ +0.14 [+0.09, +0.18]

## DM_host vs M*, binned by z



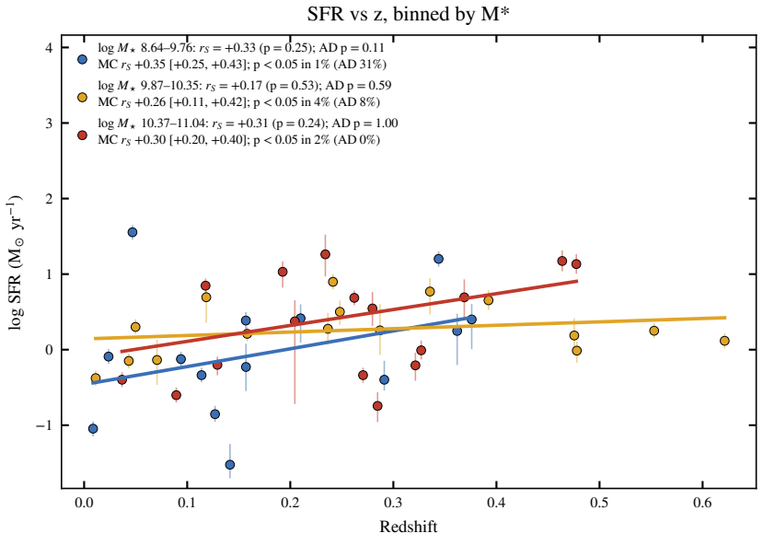

all 51: $r_S$ = -0.01 (p = 0.96); AD p = 0.91; partial $\tau$ | z = -0.01 (p = 0.91)
pooled within tertiles: $r$ = -0.02 (p = 0.88); tertiles agree: p = 0.064
through the errors (MC median [16, 84]%): all $r_S$ -0.04 [-0.12, +0.05], p < 0.05 in 0% (AD 1%)
MC partial $\tau$ -0.03 [-0.08, +0.03]

## DM_host vs SFR, binned by z



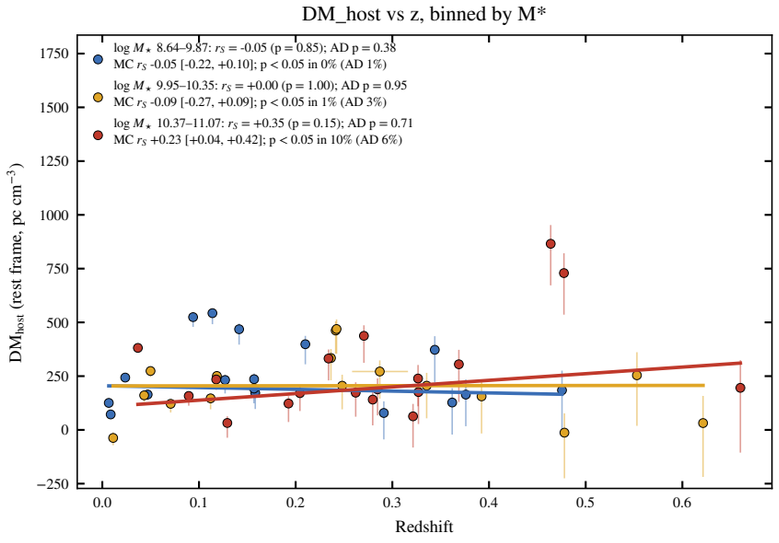

all 46: $r_S$ = +0.24 (p = 0.10); AD p = 0.020; partial $\tau$ | z = +0.19 (p = 0.073)
pooled within tertiles: $r$ = +0.40 (p = 0.013); tertiles agree: p = 0.004
through the errors (MC median [16, 84]%): all $r_S$ +0.17 [+0.08, +0.25], p < 0.05 in 7% (AD 9%)
MC partial $\tau$ +0.13 [+0.06, +0.19]

## SFR vs z, binned by M*



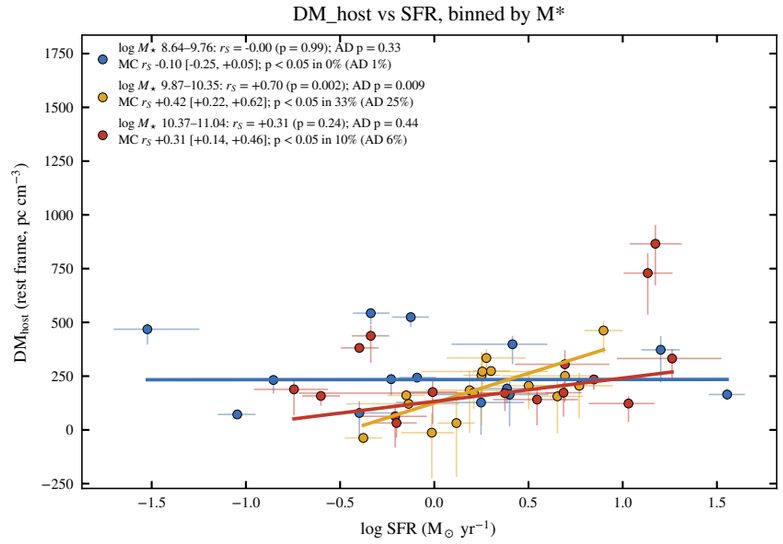

all 46: $r_S$ = +0.34 (p = 0.020); AD p = 0.10; partial $\tau$ | log $M_\star$ = +0.20 (p = 0.089)
pooled within tertiles: $r$ = +0.27 (p = 0.10); tertiles agree: p = 0.90
through the errors (MC median [16, 84]%): all $r_S$ +0.35 [+0.31, +0.38], p < 0.05 in 92% (AD 29%)
MC partial $\tau$ +0.21 [+0.18, +0.24]

## DM_host vs z, binned by M*



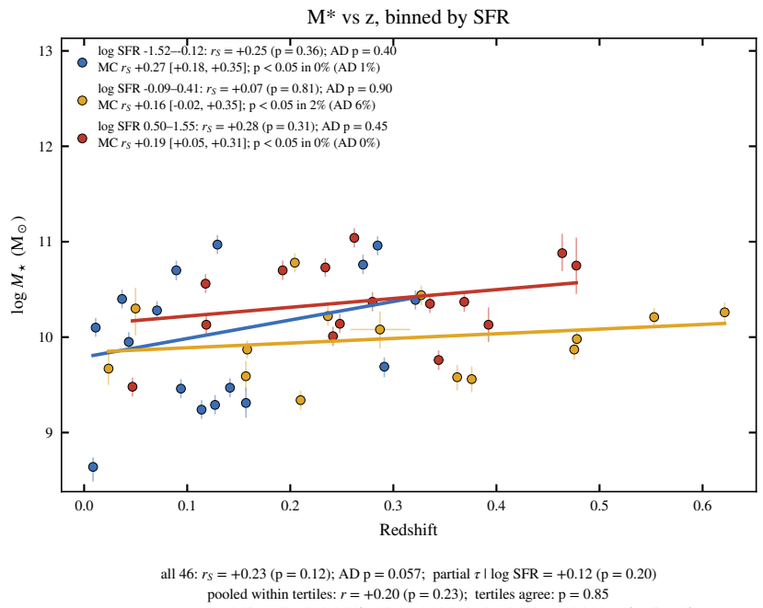

all 51: $r_S$ = +0.06 (p = 0.66); AD p = 0.95; partial $\tau$ | log $M_\star$ = +0.04 (p = 0.69)
pooled within tertiles: $r$ = +0.11 (p = 0.47); tertiles agree: p = 0.50
through the errors (MC median [16, 84]%): all $r_S$ +0.01 [-0.08, +0.09], p < 0.05 in 0% (AD 0%)
MC partial $\tau$ +0.02 [-0.04, +0.07]

## DM_host vs SFR, binned by M*



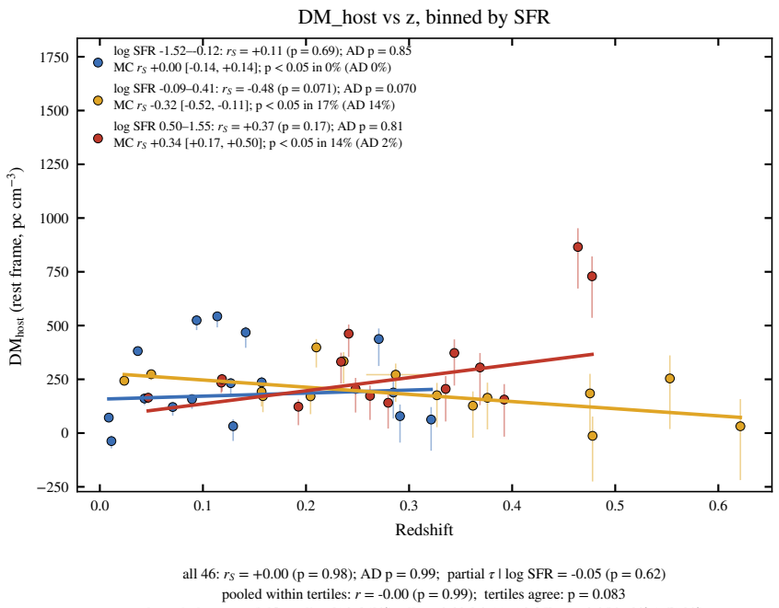

all 46: $r_S$ = +0.24 (p = 0.10); AD p = 0.023; partial $\tau$ | log $M_\star$ = +0.19 (p = 0.068)
pooled within tertiles: $r$ = +0.40 (p = 0.013); tertiles agree: p = 0.11
through the errors (MC median [16, 84]%): all $r_S$ +0.17 [+0.08, +0.25], p < 0.05 in 7% (AD 9%)
MC partial $\tau$ +0.12 [+0.06, +0.19]

## M* vs z, binned by SFR



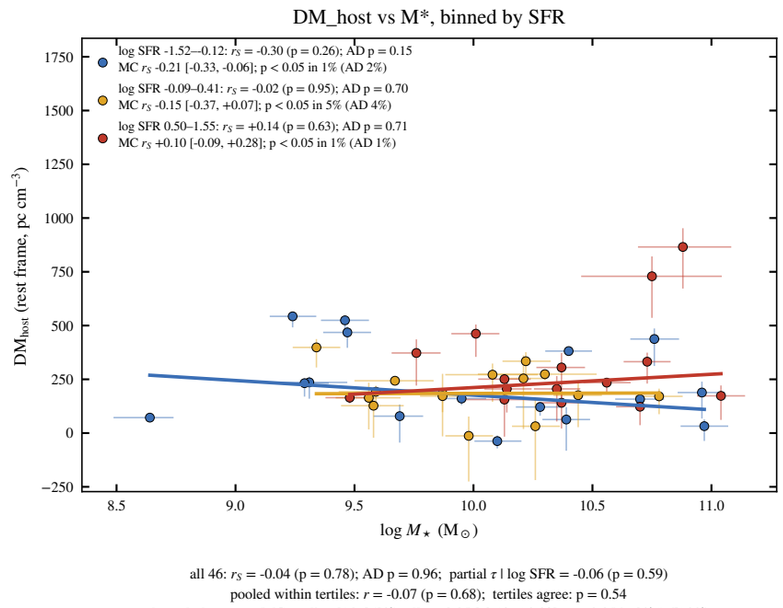

all 46: $r_S$ = +0.23 (p = 0.12); AD p = 0.057; partial $\tau$ | log SFR = +0.12 (p = 0.20)
pooled within tertiles: $r$ = +0.20 (p = 0.23); tertiles agree: p = 0.85
through the errors (MC median [16, 84]%): all $r_S$ +0.23 [+0.19, +0.27], p < 0.05 in 4% (AD 41%)
MC partial $\tau$ +0.12 [+0.09, +0.14]

## DM_host vs z, binned by SFR



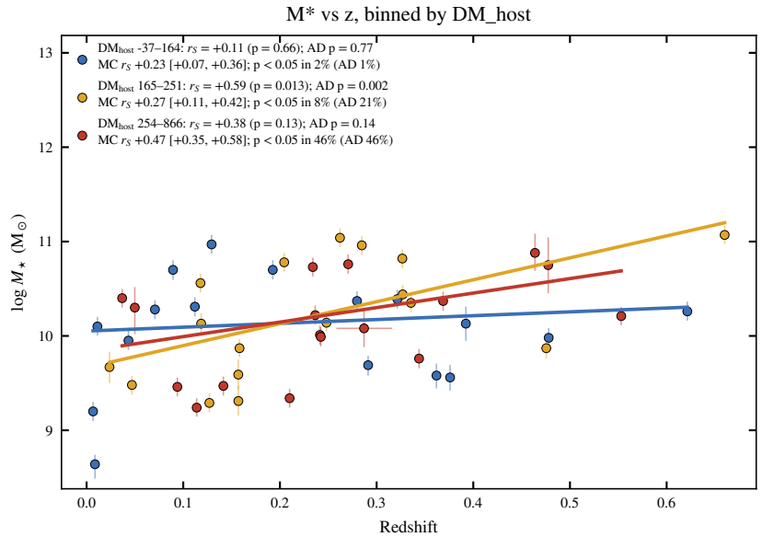

all 46: $r_S$ = +0.00 (p = 0.98); AD p = 0.99; partial $\tau$ | log SFR = -0.05 (p = 0.62)
pooled within tertiles: $r$ = -0.00 (p = 0.99); tertiles agree: p = 0.083
through the errors (MC median [16, 84]%): all $r_S$ -0.02 [-0.11, +0.05], p < 0.05 in 0% (AD 0%)
MC partial $\tau$ -0.05 [-0.11, +0.01]

## DM_host vs M*, binned by SFR



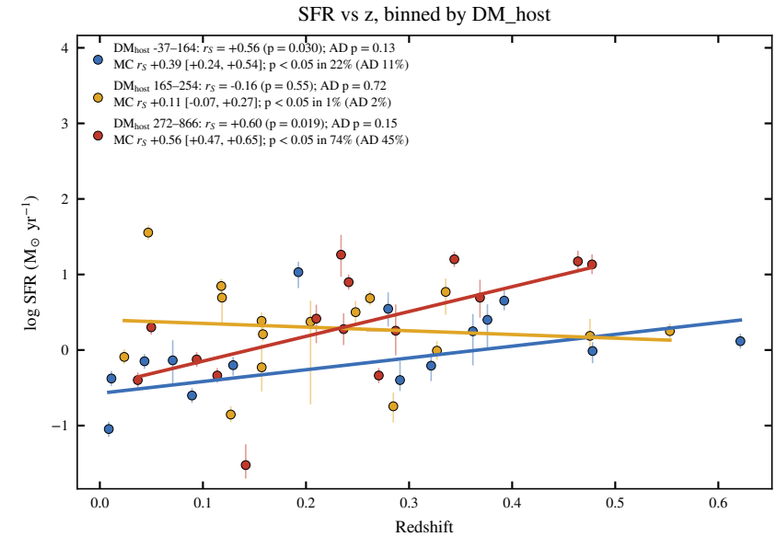

all 46: $r_S$ = -0.04 (p = 0.78); AD p = 0.96; partial $\tau$ | log SFR = -0.06 (p = 0.59)
pooled within tertiles: $r$ = -0.07 (p = 0.68); tertiles agree: p = 0.54
through the errors (MC median [16, 84]%): all $r_S$ -0.05 [-0.13, +0.03], p < 0.05 in 0% (AD 0%)
MC partial $\tau$ -0.06 [-0.11, -0.00]

## M* vs z, binned by DM_host



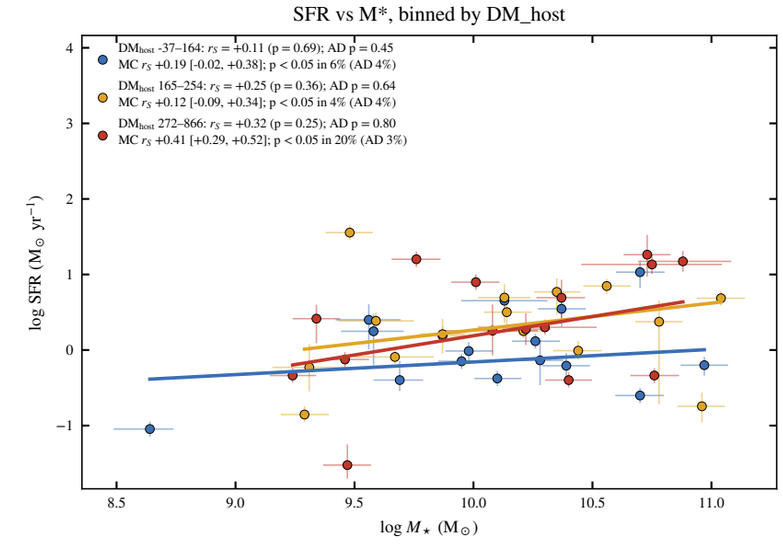

all 51: $r_S$ = +0.32 (p = 0.024); AD p = 0.023; partial $\tau$ | $DM_{host}$ = +0.22 (p = 0.034)
pooled within tertiles: $r$ = +0.38 (p = 0.012); tertiles agree: p = 0.36
through the errors (MC median [16, 84]%): all $r_S$ +0.31 [+0.27, +0.34], p < 0.05 in 82% (AD 88%)
MC partial $\tau$ +0.21 [+0.19, +0.24]

## SFR vs z, binned by DM_host




all 46: $r_S$ = +0.34 (p = 0.020); AD p = 0.097; partial $\tau$ | $DM_{host}$ = +0.23 (p = 0.025)
pooled within tertiles: $r$ = +0.36 (p = 0.028); tertiles agree: p = 0.066
through the errors (MC median [16, 84]%): all $r_S$ +0.35 [+0.31, +0.38], p < 0.05 in 92% (AD 29%)
MC partial $\tau$ +0.23 [+0.21, +0.26]

## SFR vs M*, binned by DM_host




all 46: $r_S$ = +0.25 (p = 0.097); AD p = 0.41; partial $\tau$ | $DM_{host}$ = +0.19 (p = 0.096)
pooled within tertiles: $r$ = +0.23 (p = 0.17); tertiles agree: p = 0.87
through the errors (MC median [16, 84]%): all $r_S$ +0.23 [+0.17, +0.28], p < 0.05 in 13% (AD 1%)
MC partial $\tau$ +0.17 [+0.13, +0.21]

How to read each legend entry: tertile range of the binning quantity; Spearman rank correlation $r_S$ of the plotted pair in that tertile, with its two-sided p-value (null hypothesis: $r_S = 0$).
AD: two-sample Anderson–Darling p-value that the y quantity has the same distribution for hosts below and above the median of the x quantity (5,000 permutations); it also detects non-monotonic dependence.
MC $r_S$: the same Spearman coefficient recomputed in 2,000 draws in which every value is resampled within its error bars and the tertiles are re-formed; median [16th, 84th percentile] of $r_S$ (not a p-value), and the fraction of draws with Spearman p < 0.05 (in brackets: with AD p < 0.05).